

BASIC PYTHON EXERCISE

COMM2708 Digital and Data Journalism

Week 6 Basic Python and Jupyter Notebook

Guidelines

Complete the exercise in order. Run each code cell by clicking the arrow button. Use Markdown cells to explain what your code has. The exercise moves from basic printing and variables to operators, lists, loops, methods, and functions.

Part 1 Python Basic

1. Create a Python command that prints the sentence:
Melbourne records its warmest day this month
2. Create the following variables:
 - city = "Melbourne"
 - temperature = 32
 - year = 2026Then, do these steps:
 - a. Print each variable
 - b. Create one sentence that uses all three variables
Example of the sentence: Melbourne recorded a temperature of 32 degrees in 2026.
3. A newsroom published 80 stories in January and 100 stories in February.
 - a. Calculate the difference between February and January
 - b. Calculate the total number of stories published across both months
 - c. Calculate the percent increase from Jan to Feb
4. Predict the result before running each line. Then, running the code and providing the answer from Python.
 - a. `50 > 30`
 - b. `100 == 90`
 - c. `25 != 20`
 - d. `10 <= 30`
5. Imagine these are the five most-read topics on a news website:
Topics = ["Politics", "Climate", "Business", "Health", "Sport"]
 - a. Print the first topic
 - b. Print the third topic
 - c. Print the last topic

6. Using the same topic list, write a loop that prints every topic one by one.

7. You have this list of news outlets:

News_outlets = ["abc", "the guardian", "the age", "sbs"]

- a. Use a loop and `.upper()` to print every outlet in capital letters.
- b. Replace `.upper()` with `.title()`. What changes in the output?

8. Create a function called `double()` that takes one number and multiplies it by two.

9. You have this list:

Audience = [100, 200, 300, 400, 500]

Create a `double()` function and use a loop to apply it to every number in the list.

10. A journalist has three assessment scores for his or her work:

- a. Data collection = 80
- b. Data analysis = 70
- c. Storytelling = 90

Create a function called `journalism_score()` that sums the three scores and calculates their average.

11. A news website had 125,000 visitors in January and 150,000 visitors in February. Create a function called `percent_change()` that calculates the percent change.